

Comparative Analysis of Web Interfaces Based on Human Interface Laws and Design Patterns

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This report presents an academic comparative analysis of two supermarket-related websites: Tatylu (<https://supermarkettatylu.onrender.com/>) and Supermaxi (<https://www.supermaxi.com/>). The objective is to evaluate both interfaces using four well-established human-computer interaction (HCI) laws and to identify the design patterns currently implemented in each system.

1. Jakob's Law (Familiarity and User Expectations)

Jakob's Law states that users prefer interfaces that resemble those they already know.

Tatylu presents a minimal interface with basic navigation options. While simple, it lacks common e-commerce patterns such as shopping carts, user accounts, and structured product browsing, which reduces familiarity.

Supermaxi follows standard e-commerce conventions, including categorized navigation, promotions, and shopping-related features, aligning closely with user expectations.

2. Hick's Law (Decision-Making and Choice Complexity)

Hick's Law explains that decision time increases with the number of choices unless they are properly organized.

Tatylu limits options but lacks hierarchical categorization, which may hinder efficient decision-making.

Supermaxi offers many choices but organizes them into logical categories and submenus, reducing cognitive overload.

3. Fitts's Law (Ease of Interaction)

Fitts's Law indicates that interaction efficiency depends on the size and position of clickable elements.

Tatylu uses relatively small and uniform interactive elements, making key actions less prominent.

Supermaxi implements large, visually distinct buttons and banners that facilitate faster user interaction.

4. Gestalt Principles (Visual Grouping and Organization)

Gestalt principles explain how users perceive grouped elements.

Tatylu shows limited visual grouping, making content scanning less efficient.

Supermaxi clearly separates content using visual hierarchy, color, and spacing.

5. Design Patterns Analysis

In addition to usability laws, both websites implement different interface design patterns.

Tatylu applies the following patterns:

- **Static Navigation Menu:** Simple top-level navigation suitable for small-scale websites.
- **Single-Page Layout Pattern:** Linear content presentation that reduces complexity but limits scalability.
- **Informational Design Pattern:** Focus on presenting information rather than supporting complex user transactions.

Supermaxi applies more advanced design patterns:

- Mega Menu Pattern: Supports exploration of large product catalogs.
- Card-Based Layout Pattern: Products and promotions are displayed using reusable visual cards.
- Search-Centered Design Pattern: A prominent search bar enables rapid product discovery.
- Promotional Banner Pattern: Directs user attention to offers and campaigns.
- Progressive Disclosure Pattern: Information and options are revealed gradually to avoid overload.

Conclusion

- Based on the combined evaluation of human interface laws and design patterns, Supermaxi demonstrates a more mature and scalable user interface. Its design choices enhance usability, efficiency, and user confidence.
- Tatylu benefits from clarity and simplicity; however, incorporating additional e-commerce design patterns—such as transactional flows, product categorization, and enhanced interaction feedback—would significantly improve its usability and alignment with modern interface standards.